Appendix – Supplementary Materials

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A. Data Source and Ingestion

The dataset used in this analysis were obtained from the MOTHER Database: https://mother-db.org/search/

Download script: scripts/download_mother_nmr.sh

https://github.com/ReproAnalytics/nmr-ovarian-follicle-ml/blob/main/scripts/download_mother_nmr.sh

B. Supplementary Tables

a. Supplemental Table 1. Source information for .xml data pulled from MOTHER

	Source	Title	Organization	Author_First	Author_Last	Author_Contact
0	MDB0000530-P28-Ova-106_s55.xml	Brieño Lab: NMR Postnatal Ovary P28 (Ova106-s55)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
1	MDB0000531-P1-11-s40.xml	Brieño Lab: NMR Postnatal Ovary P1 (Ova11-s40)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
2	MDB0000532-P15_F09OV-s83.xml	Brieño Lab: NMR Postnatal Ovary P15 (F09-s83)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
3	MDB0000533-P180_6Mon-6b-s86.xml	Brieño Lab: NMR 6-Months Ovary (6B-s86)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
4	MDB0000534-P5-23-s13.xml	Brieño Lab: NMR Postnatal Ovary P5 (Ova23-s13)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
5	MDB0000535-P8-F02-OV-s135.xml	Brieño Lab: NMR Postnatal Ovary P8 (F02-s135)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
6	MDB0000536-P90-NMR_3Mo-3B-s96.xml	Brieño Lab: NMR Postnatal Ovary P90 (3B-s96)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
7	MDB0000537-2653-3y-Exsub-s19.xml	Brieño Lab: NMR Ovary 3y Ex-subordinate (2653-...	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu
8	MDB0000538-2659-3y-Sub-s109.xml	Brieño Lab: NMR Ovary 3y Subordinate (2659-s109)	University of Pittsburgh	Odei	Barreñada	barrenadatalebo@upmc.edu

b. Supplemental Table 2. Summary of Statistics for the RBG analysis of all 9 slides

Slide	R (Mean ± SD)	G (Mean ± SD)	B (Mean ± SD)
MDB0000530-P28-Ova-106_s55_reduced_tissue_hist.png	134.03 ± 96.0	136.05 ± 94.09	128.34 ± 92.29
MDB0000531-P1-11-s40_reduced_tissue_hist.png	155.55 ± 79.55	163.41 ± 75.06	143.15 ± 73.05
MDB0000532-P15_F09OV-s83_reduced_tissue_hist.png	155.41 ± 79.52	162.41 ± 72.72	140.43 ± 70.52
MDB0000533-P180_6Mon-6b-s86_reduced_tissue_hist.png	158.78 ± 78.8	165.98 ± 68.41	143.09 ± 67.41
MDB0000534-P5-23-s13_reduced_tissue_hist.png	160.96 ± 76.0	168.35 ± 65.95	143.29 ± 65.37
MDB0000535-P8-F02-OV-s135_reduced_tissue_hist.png	159.86 ± 75.07	168.34 ± 68.26	145.86 ± 67.64
MDB0000536-P90-NMR_3Mo-3B-s96_reduced_tissue_hist.png	141.1 ± 92.54	144.38 ± 87.83	129.76 ± 84.43
MDB0000537-2653-3y-Exsub-s19_reduced_tissue_hist.png	165.13 ± 73.19	174.0 ± 61.16	151.23 ± 62.66
MDB0000538-2659-3y-Sub-s109_reduced_tissue_hist.png	161.09 ± 77.0	168.59 ± 68.07	147.85 ± 68.13

c. Supplemental Table 3. Training Callbacks (Output 1)

epoch	train_loss	valid_loss	error_rate	time
0	2.689905	1.431696	0.464286	02:20

d. Supplemental Table 4. Training Callbacks (Output 2)

epoch	train_loss	valid_loss	error_rate	time
0	1.573509	0.974007	0.369048	03:18
1	1.211043	0.916476	0.345238	03:21
2	0.92782	0.853306	0.333333	03:18
3	0.733885	0.784905	0.261905	03:15
4	0.579777	0.803684	0.297619	03:16

d. Supplemental Table 5. Training Callbacks (Output 3)

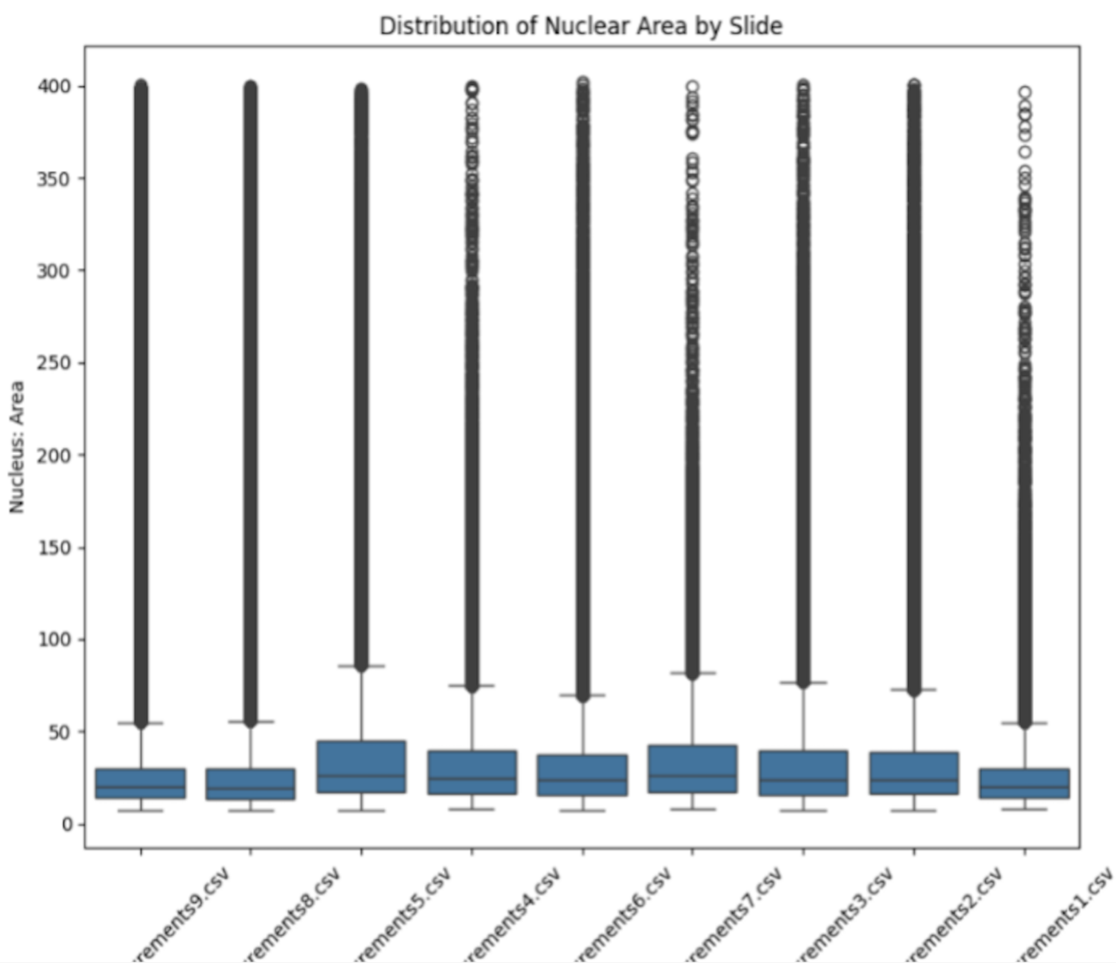
epoch	train_loss	valid_loss	error_rate	time
0	0.421417	0.783126	0.297619	03:20
1	0.439110	1.121641	0.321429	03:16
2	0.406873	1.124970	0.345238	03:17
3	0.437497	1.174675	0.321429	03:18

C. Supplemental Figures

a. Supplemental Figure 1. Cell and Nucleus Count Summary per Slide

slide	count	mean	std	median
nuclei_measurements1.csv	66253	26.335238	22.867908	19.75
nuclei_measurements2.csv	151053	35.299699	37.361070	24.00
nuclei_measurements3.csv	83598	38.149328	43.257404	23.75
nuclei_measurements4.csv	123866	33.494486	29.159414	24.50
nuclei_measurements5.csv	194773	41.546733	45.785996	26.25
nuclei_measurements6.csv	134475	33.912960	35.741969	23.50
nuclei_measurements7.csv	70463	34.742457	28.662124	26.00
nuclei_measurements8.csv	705869	27.714490	29.472896	19.25
nuclei_measurements9.csv	921344	27.328513	27.530887	19.75

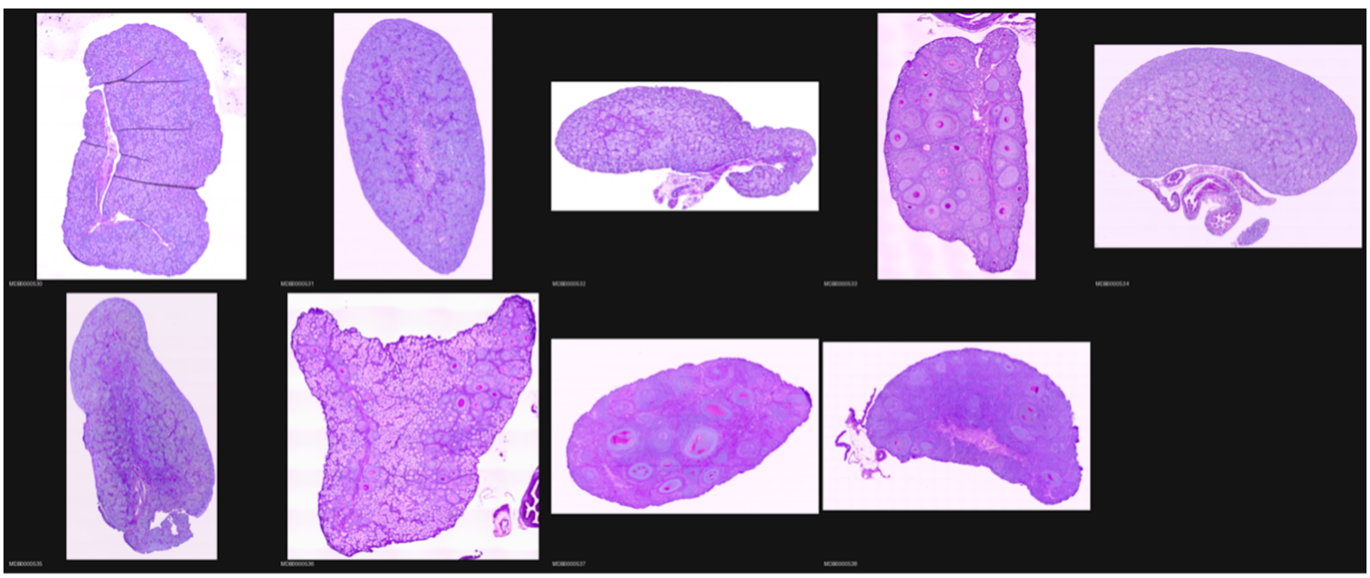
b. Supplemental Figure 2. Boxplot Representation of the Distribution of Nuclear Area Across all 9 Slides.



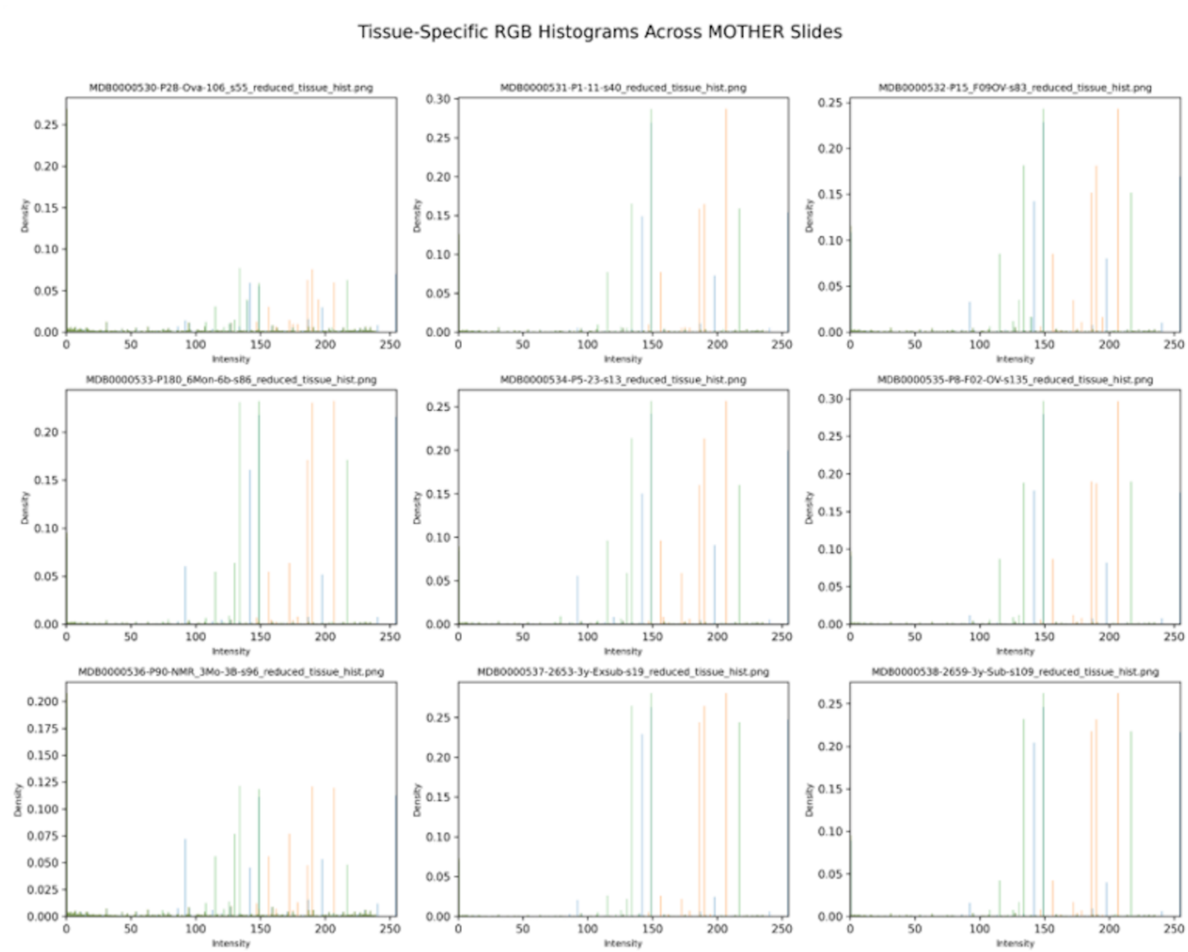
c. Supplemental Figure 3. Bootstrapping output from QuPath Analysis

```
nuclei_measurements9.csv 95% CI: [27.27250253 27.38201308]
nuclei_measurements8.csv 95% CI: [27.65097524 27.77989676]
nuclei_measurements5.csv 95% CI: [41.3428683  41.75334267]
nuclei_measurements4.csv 95% CI: [33.33797955 33.64982723]
nuclei_measurements6.csv 95% CI: [33.72407297 34.107099  ]
nuclei_measurements7.csv 95% CI: [34.52627718 34.95122804]
nuclei_measurements3.csv 95% CI: [37.85620058 38.44432044]
nuclei_measurements2.csv 95% CI: [35.1153255  35.48538489]
nuclei_measurements1.csv 95% CI: [26.17093632 26.50465875]
```

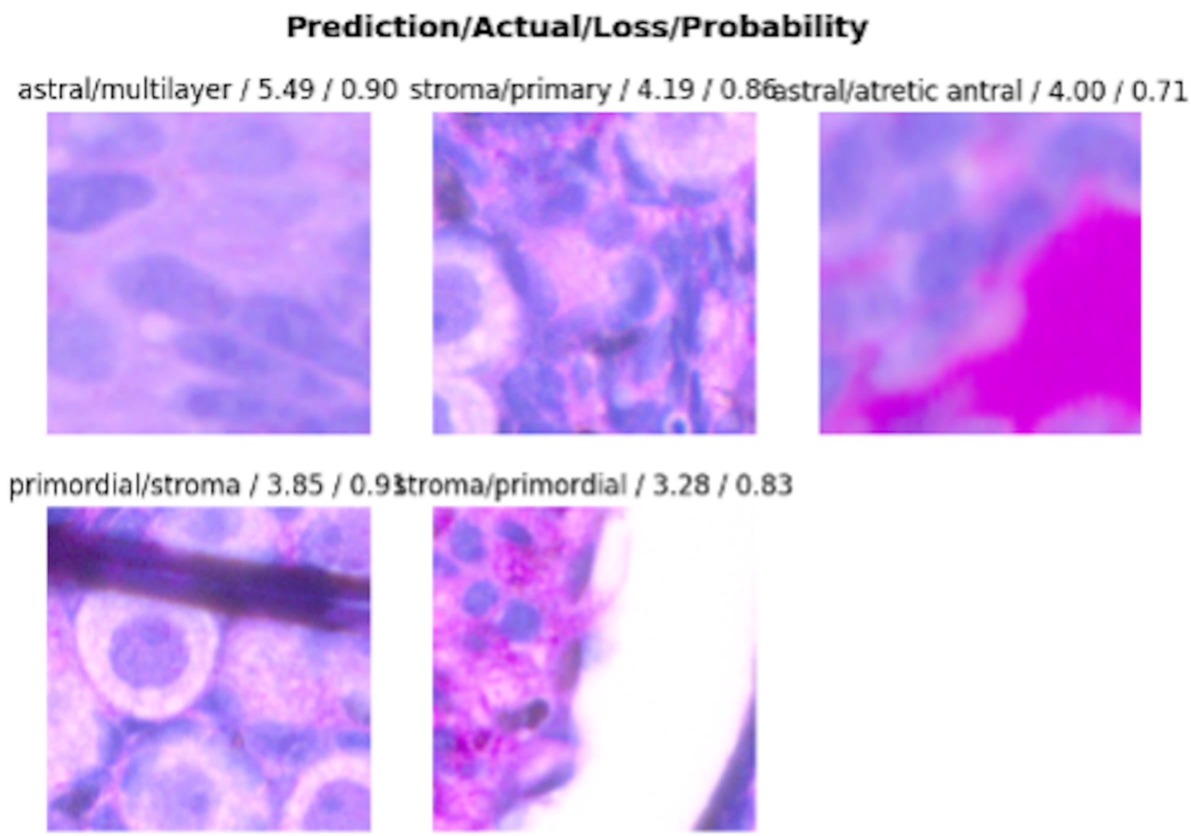
d. Supplemental Figure 4. Montage showing all 9 whole-image slides



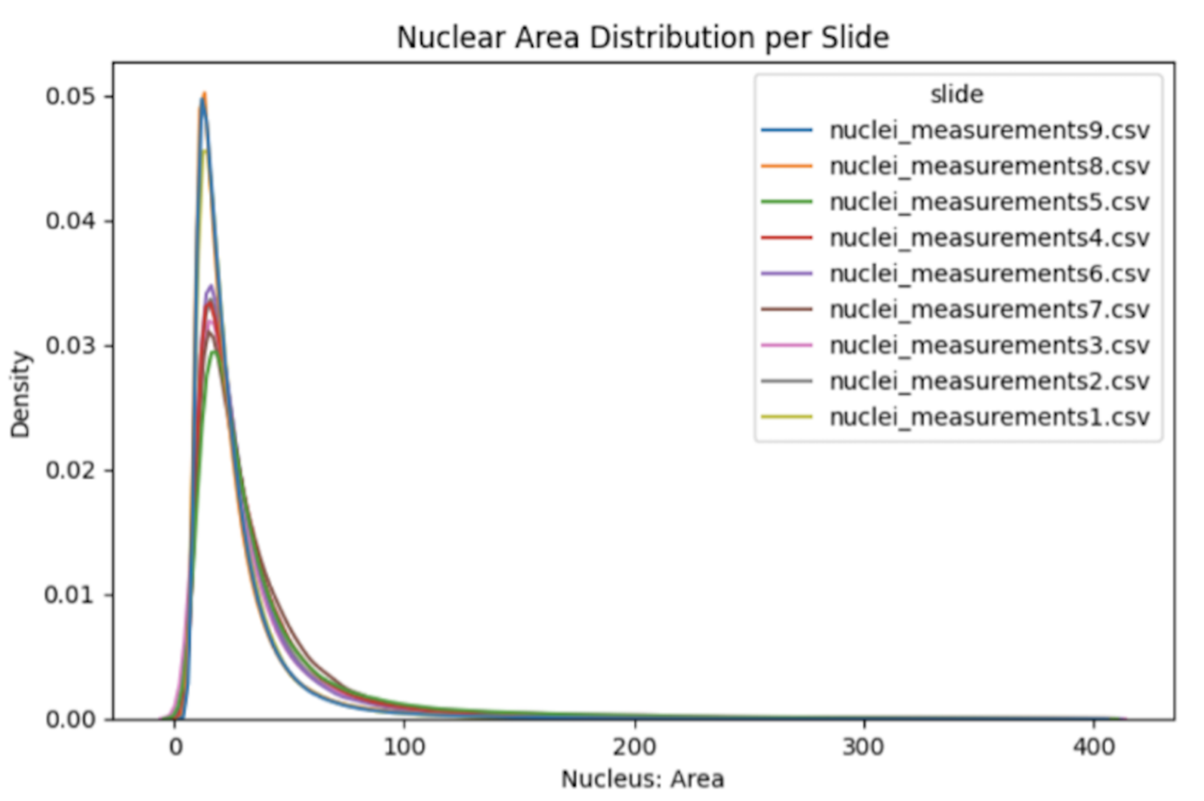
e. Supplemental Figure 5. Histograms of the intensity and density of RBG distribution per slide.



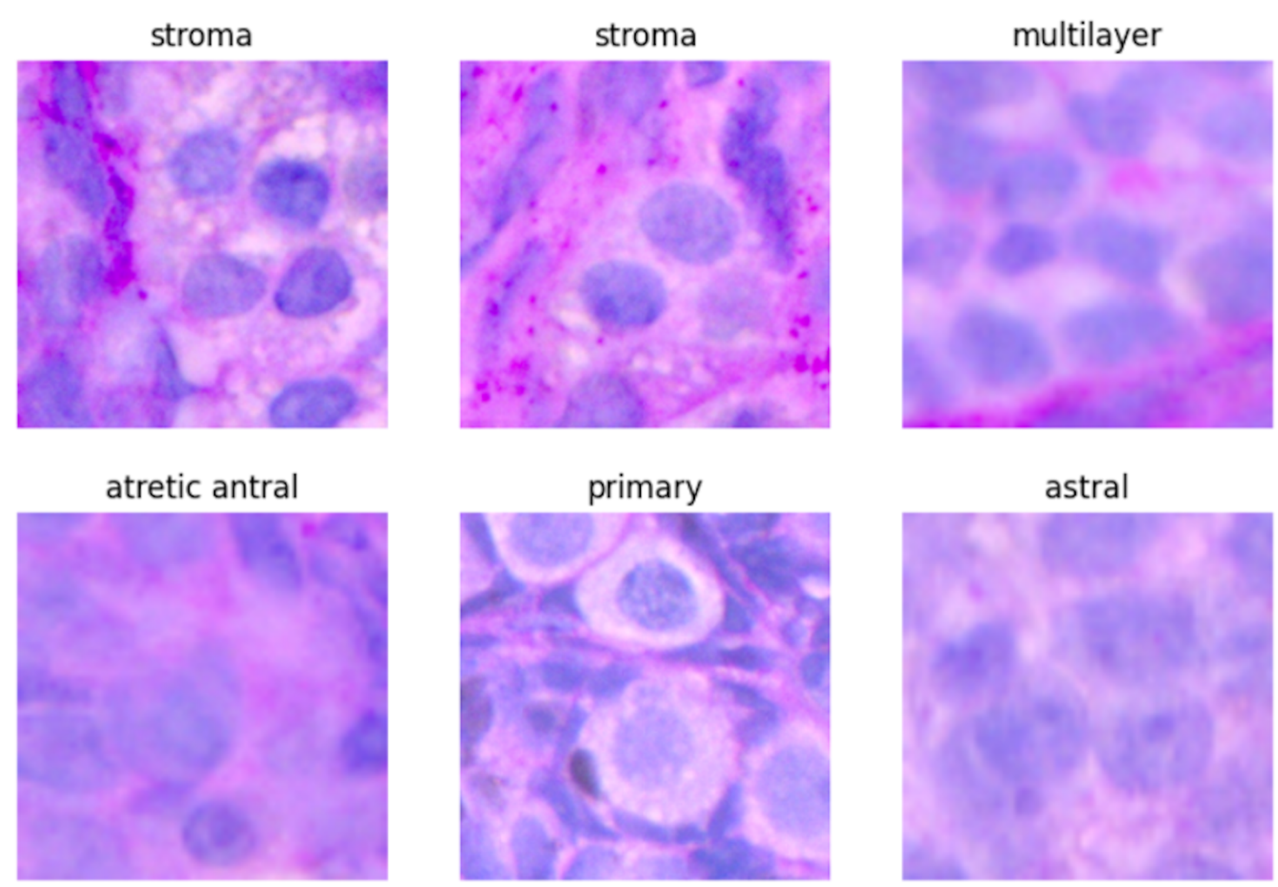
f. Supplemental Figure 6. Confusion Matrix Probability Results



g. Supplemental Figure 7. Distribution of nuclear area detected across all slides via kernel density plot.



h. Supplemental Figure 8. Classification of cell types.



D. References

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